

Vanishing Urban Wetlands

Are Cities Losing Their Natural Flood Buffers?

A survey of the causes, evidence and remedial measures related to the decline of urban wetlands and their role in flood mitigation, supported by five city case studies and a structured awareness questionnaire.

64% of global wetlands lost since 1900 (Ramsar)	5 cities surveyed across three continents	\$21T in wetland ecosystem services lost since 1970
--	---	--

SUBMISSION DETAILS

Team Member	Section / Roll No.	Registration No.
Santosh Paul	K3P26CP52	12607307
Supriya Pal	K3026CP53	12607267
Shivang Ratnam	RK3P26CPB70	12607190

Course	Details
Course	CHE110 - Environmental Studies
Session	2026-27
Report type	Internet-based survey report with questionnaire
Focus	Urban wetlands, flood buffering and disaster risk

“Wetlands are not wastelands — they are the quiet infrastructure that keeps our cities from drowning.”

Abstract

Wetlands — marshes, lakes, floodplains and mangroves — act as a city's sponge, storing monsoon and storm water and releasing it slowly. This report surveys what has happened where cities have removed them. Drawing on published research, government data and news investigations, it examines five cities on three continents (Chennai, Bengaluru, Mumbai, Jakarta and Houston), identifies the main drivers of wetland loss, reviews the policy framework, and evaluates restoration and community-led solutions. A 12-question questionnaire on awareness of urban wetlands supports the secondary research.

Main finding: in every city studied, wetland loss preceded worsening floods, and the gap between law and enforcement — not a lack of scientific knowledge — is the critical weakness. Restoration projects show that flood buffers can be rebuilt where they are protected and monitored.

Contents

- 1 Introduction and Objectives
- 2 Methodology
- 3 Understanding Wetlands and Flood Buffering
- 4 The Global and Economic Picture
- 5 Five City Case Studies
- 6 Drivers of Wetland Loss
- 7 Policy and Legal Framework
- 8 Solutions and Community Action
- 9 Awareness Questionnaire
- 10 Myths vs. Facts
- 11 Findings and Recommendations
- 12 Conclusion
- References

1 Introduction and Objectives

A wetland is any area where water covers the soil, or is present at or near the surface, for all or part of the year. Because wetlands sit between land and water, they are ecologically rich but easy to overlook in planning, and they have historically been dismissed as “wasteland” fit only for draining and building over. This report asks a pointed question: as cities pave over their wetlands, are they also paving over one of their best natural defences against flooding?

The topic connects to core themes of environmental studies: ecosystem structure and function, ecosystem goods and services, biodiversity loss, pollution, disaster management, and the policies and community movements that shape how societies manage natural resources.

Objectives of the survey

- To explain what urban wetlands are and the functions they perform, particularly flood regulation.
- To examine, using data and case studies, how wetland loss has increased flood risk in cities across India, Southeast Asia and the United States.
- To identify the principal drivers of wetland loss — encroachment, pollution and climate change — and how they interact.
- To assess the policy and legal framework governing wetland conservation in India and internationally.
- To review nature-based and community-driven solutions that are restoring wetlands and reducing flood risk.
- To measure baseline awareness of urban wetlands among respondents through a structured questionnaire.

2 Methodology

The survey combined two complementary approaches.

1. **Secondary research (internet-based).** Published research, government and intergovernmental data (Ramsar, Nature, Nature Water), academic institutions, and news investigations were reviewed for global statistics and five city case studies. Sources are listed in the References.
2. **Awareness questionnaire.** A 12-question multiple-choice Google Form, “Vanishing Urban Wetlands”, was designed in four sections: awareness, urbanisation and flooding, environment and biodiversity, and awareness and solutions (see Section 9).

Scope and limitations

- Estimates of wetland loss differ because studies use different baseline years and methods; figures are presented with their baselines.
- City statistics use different metrics (area, number of lakes, share of cover), so cross-city comparisons are indicative, not exact.
- Web sources were accessed in September 2026; figures should be verified against original publications before formal citation.

3 Understanding Wetlands and Flood Buffering

3.1 Key terms

Term	Meaning in this report
Wetland	Land where water covers the soil or stays near the surface for all or part of the year, including marshes, swamps, floodplains and mangroves.
Ecosystem services	Benefits people obtain from ecosystems: provisioning, regulating (e.g. flood control), cultural and supporting services.
Ramsar site	A wetland designated as internationally important under the 1971 Ramsar Convention.
Encroachment	Unauthorised occupation or conversion of a lake bed, marsh or floodplain for construction or other use.
Eutrophication	Excess nutrient enrichment (sewage, fertiliser runoff) that depletes oxygen and harms aquatic life.
Groundwater recharge	Percolation of surface water to replenish underground aquifers.
Blue-green infrastructure	Planning that combines water and vegetated elements (wetlands, parks, permeable surfaces) to manage water.
Disaster Risk Reduction (DRR)	Policy and practice to minimise vulnerability and disaster risk through prevention and mitigation.

3.2 Types of urban wetland and their flood role

Wetland type	Example	Primary flood-control function
Freshwater marsh	Pallikaranai Marsh, Chennai	Temporary storage of excess rainwater; slow release
Urban lake / tank	Bellandur Lake, Bengaluru	Rainwater harvesting; groundwater recharge
River floodplain	Mithi River, Mumbai	Absorbs peak river discharge during heavy rain
Mangrove / coastal wetland	Thane Creek, Mumbai; Angke, Jakarta	Dissipates storm surge and wave energy

3.3 How wetlands buffer floods

A natural catchment with wetlands, ponds and permeable ground absorbs a large share of rainfall and releases the rest gradually. When wetlands are replaced with concrete, asphalt and buildings, the same rainfall becomes

surface runoff almost immediately: it moves faster, arrives sooner and pools in whatever low-lying area is left. Unlike a storm drain, which only moves water, a wetland stores it.

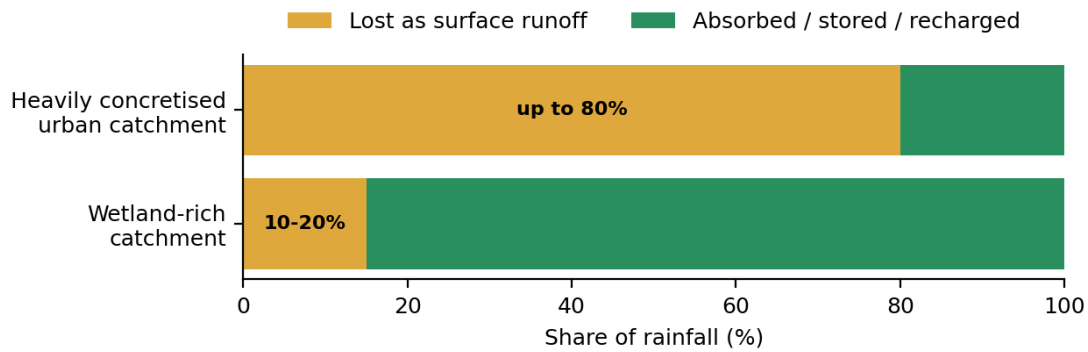


Fig. 1 — Approximate share of rainfall lost to runoff in natural and concretised catchments.

4 The Global and Economic Picture

4.1 How much has been lost?

Estimates vary with baseline year and methodology, but every credible study agrees on the direction: wetlands are disappearing faster than most other ecosystems.

Source	Baseline	Estimated loss
Ramsar Wetlands Extent Index	1970–2008 (sampled sites)	~40%
Global Wetland Outlook 2025	Since 1970	22% (411 million ha)
Nature (2023) global reconstruction	Since 1700	21–35%
Davidson (2014) literature review	Since 1900	54–57% (average)

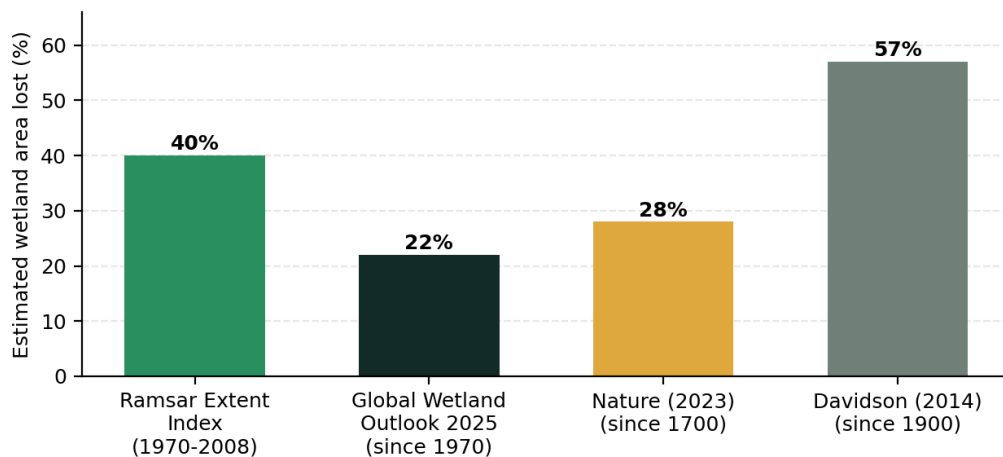


Fig. 2 — Global wetland loss estimates by baseline year and study.

Two patterns stand out. Loss has been faster in Asia than in Europe or North America, where rates have slowed since the 1980s. Urban and rural development are also a major direct cause: in the United States they account for just over a third of documented wetland losses.

4.2 Biodiversity cost

Wetlands are disproportionately rich relative to their area, supporting fish, amphibians, and resident and migratory birds. Pallikaranai Marsh alone has several hundred recorded flora and fauna species, over 100 bird species (including migrants on the Central Asian flyway) and about 20 reptile species. In Bengaluru, waterbird populations have declined with lake and reedbed loss. A wetland's biodiversity and its water-storage capacity depend on the same intact vegetation and hydrology, so habitat loss and flood risk are one story.

4.3 Economics of a wetland

Study / source	Value (US\$ per hectare per year)
Land converted to agriculture (lost value)	~\$3,200
Intact freshwater marsh (Millennium Ecosystem Assessment)	~\$7,700
US average flood-mitigation value (Nature Water, 2025)	~\$15,738 (up to \$300,000+ in high-risk areas)
Global inland marsh, median ecosystem-service value	~\$73,000

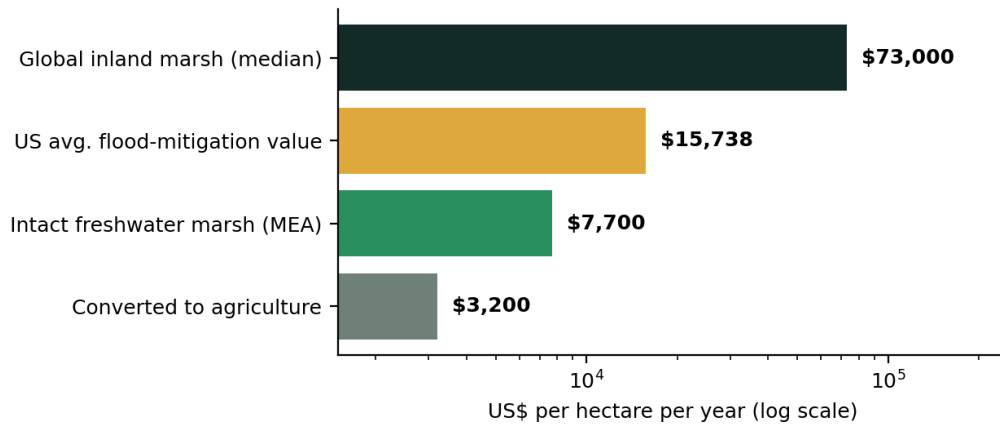


Fig. 3 — An intact wetland generates far more annual value than the land does once converted.

\$10.1B

added U.S. flood-insurance claims since 1985 attributed to wetland loss (Nature Water, 2025)

\$21T

wetland ecosystem services lost globally since 1970

Cities that fill in wetlands give up a recurring economic asset in exchange for a one-time gain in buildable land.

5 Five City Case Studies

The five cities span three continents and very different climates, yet each tells a version of the same story: wetlands removed, flood risk increased, and the human cost measured in lives and billions of dollars.

5.1 Chennai, India — Pallikaranai Marsh

- Historic extent ~5,000–7,000 ha (1960s) has fallen to ~690 ha of protected Reserve Forest — about 90% lost.
- City-wide water-body area fell from 12.6 sq.km (1893) to 3.2 sq.km (2017).
- December 2015 floods: Velachery, built on former marshland, was among the worst-hit areas.
- An estimated 1,500–2,000 tonnes of waste are dumped daily, with a municipal dump yard inside the marsh.
- A 2025 study found 165 water bodies (31 lakes, 134 tanks) had vanished from the wider catchment, about 7.35 sq.km of storage. Ramsar status since 2022.

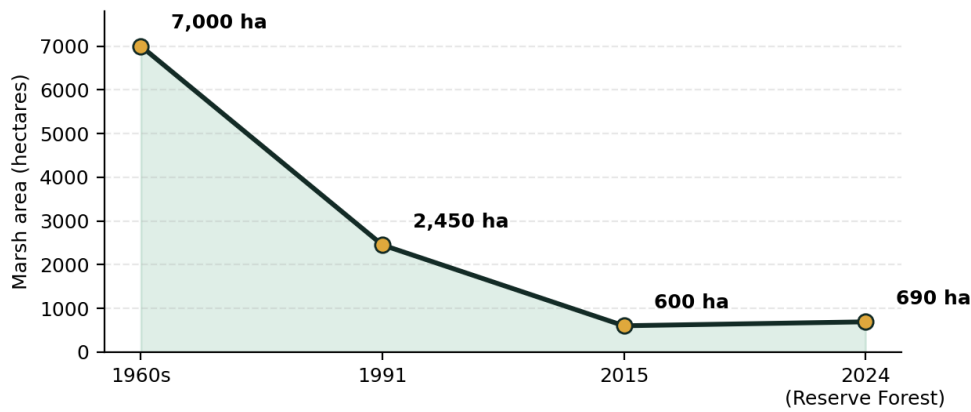


Fig. 4 — Pallikaranai Marsh has lost roughly 90% of its historical extent.

5.2 Bengaluru, India — The vanishing lakes

- Functioning lakes fell from 209 (1973) to 93 (2011); open water from 3.4% of city area (1973) to under 1% (2013).
- A 2023 survey found 20% of the combined area of 837 lakes encroached; only 91 were encroachment-free.
- Neighbourhoods built on lake beds or blocked rajakaluves (storm-water canals) flood repeatedly.

5.3 Mumbai, India — Mangroves and the 2005 deluge

- About 40% of wetland and mangrove cover lost between the early 1990s and 2005.
- 26 July 2005: 944 mm of rain in 24 hours; over 1,000 deaths; the city was paralysed for days.
- Construction on the Mithi River floodplain, including the Bandra-Kurla Complex on reclaimed mangrove land, contributed; a high tide blocked drainage outfalls. Flood-prone zones have grown ~18% since 2005.

5.4 Jakarta, Indonesia — The sinking megacity

- About 97% of natural wetlands converted to urban use; green and wetland cover fell from 35% (1965) to under 10%.
- 93% of the Angke mangrove forest (272 of 291 ha) destroyed; land subsiding up to 25 cm a year in northern districts.
- January 2020 floods killed 60+ people and displaced tens of thousands.

5.5 Houston, USA — Hurricane Harvey's warning

- About 25,000 acres (~10,000 ha) of wetlands lost region-wide, 1992–2010; White Oak Bayou watershed lost 70%.
- Lost flood storage of ~4 billion gallons, valued at ~\$600 million.
- Hurricane Harvey (2017): 88 deaths, \$198 billion in damages, 250,000+ structures flooded. Much loss came through many small permitted developments — “death by a thousand cuts.”

5.6 Comparison

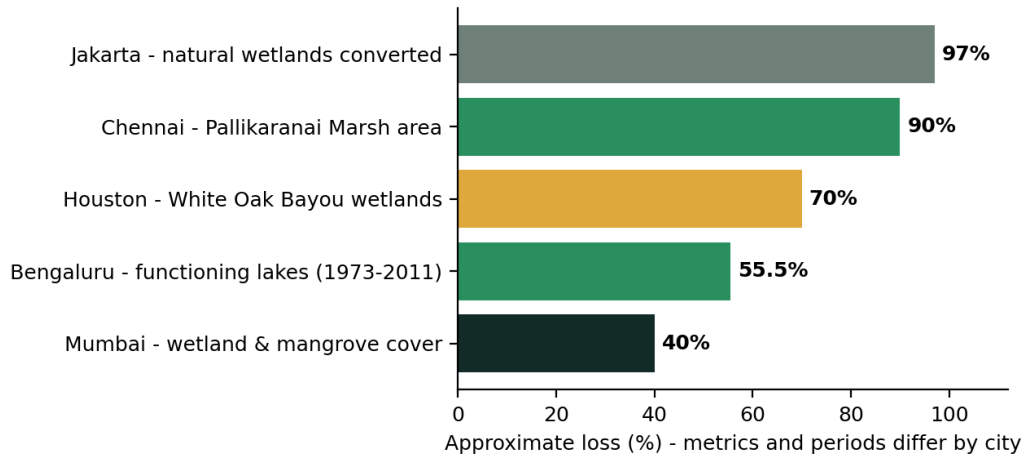


Fig. 5 — Approximate wetland loss in the five cities (metrics and periods differ; Bengaluru figure derived from 209 → 93 lakes).

City	Key loss	Key flood event	Present response
Chennai	~90% of marsh	Dec 2015 floods	Ramsar (2022); restoration planned
Bengaluru	116 of 209 lakes lost	Repeated lake-bed flooding	Citizen lake groups; encroachment surveys
Mumbai	~40% wetland/mangrove	26 July 2005; 1,000+ deaths	Community mangrove replanting
Jakarta	~97% of wetlands	Jan 2020; 60+ deaths	Subsidence and drainage debate
Houston	70% (White Oak Bayou)	Harvey 2017; \$198B	Buffalo Bayou Park restoration

6 Drivers of Wetland Loss

Encroachment and unplanned growth. The single biggest driver in the cities surveyed: conversion of lake beds, marshes and floodplains into housing, IT parks, roads and “reclaimed” commercial land. Chennai’s IT corridor on Old Mahabalipuram Road alone is estimated to have consumed around 230 sq.km of marshland.

Pollution and solid-waste dumping. Sewage, industrial effluent and waste clog channels, reduce water-holding capacity and kill vegetation that filters floodwater; dumping also contaminates groundwater.

Climate change. Warmer air holds more moisture, so rainfall increasingly arrives in short, intense bursts — the kind a paved catchment handles worst.

Driver	Mechanism	Clearest example
Encroachment	Removes storage permanently; blocks drainage paths	Bengaluru lake-bed layouts; Chennai IT corridor
Pollution / dumping	Clogs channels; kills vegetation; reduces capacity	Pallikaranai (1,500–2,000 tonnes/day)
Climate change	Intensifies short-duration rainfall beyond design capacity	Mumbai’s swings between drought and flood
Land subsidence	Sinks land below sea level, trapping floodwater	Jakarta (up to 25 cm/year)
Permitted incremental development	Many small approvals cumulatively erase capacity	Houston’s White Oak Bayou (70% lost)

7 Policy and Legal Framework

Internationally, wetland protection is anchored by the Ramsar Convention (1971), which commits signatories to conserve and make “wise use” of wetlands of international importance. India is a signatory with over 90 Ramsar sites, the most in South Asia.

Year	Instrument	Key provision
1971	Ramsar Convention	Commits signatories to conserve wetlands of international importance
2002	East Kolkata Wetlands — Ramsar	Recognised for natural sewage treatment and fish farming
2017	Wetlands (Conservation & Management) Rules	State inventories, conservation plans, State Wetland Authorities; bans unapproved industrial use, encroachment and dumping
2022	Pallikaranai Marsh — Ramsar	International recognition of ecological value and vulnerability
2023	Amrit Dharohar Scheme	Strengthens conservation and livelihoods at Ramsar sites
Dec 2024	Supreme Court of India order	Directs states to ground-truth ~30,000 unmapped wetlands under Article 21

The consistent criticism is not that rules are absent but that enforcement is inconsistent: state authorities are under-resourced, inventories go unmapped for years, and encroachment continues even inside notified boundaries, as at Pallikaranai. In disaster-management terms (NDMA), wetland protection belongs in prevention (legal protection of boundaries), mitigation (restoring storage before the monsoon), preparedness (mapping former-wetland flood zones) and recovery.

8 Solutions and Community Action

8.1 Restoration that works

Location	Intervention	Documented result
Wuhan, China	389 “sponge city” projects across 38.5 sq.km	Reduced flooding; summer temperatures down 3°C in project areas
Jinhua, China	Yanweizhou Park: flood wall replaced by terraced wetland	Floodwater drains quickly instead of pooling
Haikou, China	Mangrove and terraced wetland restoration, Meishe River	Better water quality; inundation largely stopped
Chennai, India	World Bank-funded restoration of Pallikaranai (proposed)	Aims to restore marsh function
Mumbai, India	Community-led mangrove replanting	Slow but steady recovery of coastal buffers
Houston, USA	Buffalo Bayou Park (\$58M public-private)	Riparian restoration, flood detention and amenity

8.2 Community and environmental movements

India has a long tradition of grassroots action that wetland conservation now follows: the Bishnoi movement (1730s, Rajasthan), Chipko (1973), Appiko (1983), Silent Valley (1978–81), Narmada Bachao Andolan (1985–present) and the Tehri Dam conflict. In Bengaluru and Chennai, citizen groups conducting lake walks, documenting encroachments and petitioning courts were directly responsible for forcing government action, including Ramsar recognition of Pallikaranai. Legal protection alone rarely suffices without sustained public participation and monitoring.

9 Awareness Questionnaire

To complement the secondary research, a 12-question multiple-choice Google Form was circulated. It tests awareness, understanding of the flood link, and willingness to support protection. Knowledge questions (Q2–Q8, Q10, Q11) have an expected answer based on the evidence in this report; Q1, Q9 and Q12 capture awareness and opinion.

No.	Section	Question (short form)	Expected answer
Q1	Awareness	Had you heard of an “urban wetland” before?	Opinion / awareness

No.	Section	Question (short form)	Expected answer
Q2	Awareness	Which can be considered a wetland?	All of the above
Q3	Awareness	Benefit of wetlands in heavy rainfall	Store excess water temporarily
Q4	Urbanisation	Effect of filling or building over wetlands	All of the above
Q5	Urbanisation	Why loss makes urban flooding harder to manage	Less natural space to hold water
Q6	Urbanisation	Activity most likely to threaten wetlands	Construction and land conversion
Q7	Biodiversity	Why wetlands matter for biodiversity	Habitat and food for many organisms
Q8	Biodiversity	Example of an ecosystem service	All of the above
Q9	Biodiversity	Biggest reason wetlands are disappearing	Opinion (report: encroachment)
Q10	Solutions	Approach that helps protect wetlands	All of the above
Q11	Solutions	Who has a role in protecting wetlands	Everyone listed
Q12	Solutions	Likelihood of supporting protection after learning	Opinion / attitude

Response summary (to be completed from the Google Form)

Measure	Total respondents (n)	Number	Percentage
Had heard of “urban wetland” (Q1: Yes)	_____	_____	_____ %
Answered Q2–Q8, Q10–Q11 correctly (average)	_____	_____	_____ %
Named construction as biggest threat (Q6)	_____	_____	_____ %
Would “Definitely” or “Probably” support protection (Q12)	_____	_____	_____ %

Interpretation to be added once responses are tallied: compare awareness (Q1) with knowledge scores, and check whether respondents’ own view of the biggest threat (Q9) matches the evidence that encroachment is the leading driver.

10 Myths vs. Facts

Myth	Fact
“Wetlands are just wastelands.”	Among the most productive ecosystems on Earth; an estimated \$21 trillion in wetland services has been lost since 1970.
“Filling a small lake is harmless.”	Small losses compound: White Oak Bayou lost 70% of its wetlands through many individually small permits.
“Flooding is caused only by heavy rain.”	Rain is the trigger, but wetland loss removes the buffer; a paved catchment can lose up to 80% of rainfall to runoff.
“Restoration doesn’t really work.”	Wuhan’s sponge-city projects and other restorations show measurable flood reduction.
“Only poor or developing cities are affected.”	Houston and Jakarta show the same pattern at very different income levels.

11 Findings and Recommendations

Key findings

1. Wetland loss and flood risk are linked in every city studied, across three continents and different income levels.

2. Encroachment is the largest driver, amplified by waste dumping and increasingly intense rainfall.
3. Intact wetlands are worth several times more per hectare each year than the converted land, before flood damage is counted.
4. Laws exist; enforcement and mapping are the gap.
5. Restoration and citizen monitoring measurably work.

Recommendations

- **Map and notify:** Complete and publish wetland inventories and demarcate boundaries on the ground.
- **Plan with water:** Treat wetlands as flood-control infrastructure in city master plans; require storage compensation for any approved fill.
- **Enforce:** Stop dumping and sewage inflow, and fund State Wetland Authorities adequately.
- **Restore:** Adopt sponge-city and blue-green designs, and restore former wetlands before the monsoon season.
- **Involve citizens:** Support citizen lake walks, encroachment reporting and school and college awareness programmes.

12 Conclusion

The evidence points to one consistent conclusion: cities that pave over their wetlands dismantle, piece by piece, one of the cheapest and most effective flood-control systems available to them. Chennai's shrunken marsh, Bengaluru's vanishing lakes, Mumbai's depleted mangroves, Jakarta's near-total wetland loss and Houston's watershed development each carry the same lesson, and each city has paid with worsening, recurring floods.

The evidence for solutions is just as concrete. The core challenge is closing the gap between what rules say on paper and what happens to a lake bed once a bulldozer arrives. That gap will decide whether cities keep their natural flood buffers or keep losing them.

References

- [1] Ramsar Convention Secretariat — "Wetlands: A Global Disappearing Act" factsheet; Global Wetland Outlook 2025. [ramsar.org](https://www.ramsar.org)
- [2] Fluet-Chouinard, E. et al. (2023). Global wetland reconstruction study, Nature; via McGill University Newsroom and World Economic Forum.
- [3] Davidson, N. C. (2014). Literature review of global wetland loss, Marine and Freshwater Research.
- [4] IISc CES / B.First — "Decaying Lakes of Bengaluru"; India Water Portal — "Bengaluru Lakes, in Crisis!"; Deccan Herald — Bengaluru lake-encroachment survey (2023).
- [5] WWF India "Water Tales"; Citizen Matters, Madras Musings, The Quint — Pallikaralai Marsh reporting; Journal of Forest and Wildlife (2024).
- [6] India Water Portal, Eco-Business, Disaster.Shiksha — Mumbai 2005 floods and mangrove-loss reporting.
- [7] Next City, Mongabay, National Geographic, The Conversation, Science News Explores — Jakarta subsidence and wetland loss.
- [8] Texas A&M (Texas Sea Grant), Capital & Main, Eos/Nature Water, Lincoln Institute of Land Policy — Houston wetland loss and Hurricane Harvey.
- [9] Resources for the Future; Nature Water (2025) — economic valuation of wetland flood mitigation in the U.S.
- [10] Legal Service India, Vajiram & Ravi — summaries of India's Wetlands Rules 2017 and Ramsar sites in India.
- [11] NPR, Reasons to be Cheerful, Esri, MDPI — China's "Sponge City" programme case studies.
- [12] Web sources accessed September 2026. Figures should be verified against original publications before formal citation.